A direct small-noise comparison in the iid Berry–Esseen problem

Bartosz Kołodziejek Kacper Rodziewicz

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We prove a comparison on an explicit small-noise domain, uniform over all integers $n \geq 42$. This is one part of a possible proof of the sharp iid Berry–Esseen bound. It does not establish the full bound. Throughout, $S_n = X_1 + \cdots + X_n$ for independent copies of the indicated standardized law X, Φ is the standard normal distribution function, and $C_{\mathrm{E}} = (3 + \sqrt{10})/(6\sqrt{2\pi})$. The function F_B is the distribution function of $\mathrm{Bin}(n, p)$. The final implication uses the strict finite-n Bernoulli theorem of Schulz, *The optimal Berry–Esseen constant in the binomial case*, dissertation, Theorem 1, p. 15, German National Library copy.

AI systems assisted the derivation, checking, and writing of this note. The analytic proof, exact scalar checks, and independent polynomial review are separate artifacts; agreement between AI systems is not proof.

1 A direct comparison for small noise

The small-noise part of the comparison can be proved from the incoming 0.6 strip itself. Neither the sharper half-amplitude bound nor the fractional-part and marked-crossing estimates are needed in the following range. This is a comparison of actual conditional sum laws; contact and a predecessor bound are needed only to obtain its premises elsewhere.

Theorem 1.1. *Let $n \geq 42$, $2/5 \leq p \leq 53/100$, $q = 1 - p$, and $Y = B + U$, where $\mathbb{P}(B = 1) = p$ and $\mathbb{E}(U \mid B) = 0$. Suppose the two conditional laws satisfy*

$$|U_j| \leq 3/5, \qquad U_0 \leq a_0, \qquad U_1 \geq -a_1, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 9/125.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v + s)^{-1/2}$, $X = \ell(Y - p)$, and $\rho = \mathbb{E}|X|^3$. Assume $0 < \lambda \leq 1/500$ and $|H| \leq 46/25$. Choose an integer k and $u \in [-34/125, 91/125]$ such that $np + H/\ell = k + u$. Write $\delta = k - np$, $\rho_B = (1 - 2v)/\sqrt{v}$,

$$a = \sqrt{n}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - C_{\mathrm{E}}\rho, \quad C_B = \frac{\sqrt{n}}{\rho_B}\{F_B(k) - \Phi(\delta/\sqrt{nv})\}.$$

Then $1 \leq k \leq n - 1$ and

$$a + \rho_B(C_{\mathrm{E}} - C_B) < -\lambda/100.$$

In particular no law in this range can have $a \geq 0$, by the sharp Bernoulli bound. The theorem does not cover larger noise or the remaining probability panels.

1.1 The exact comparison and uniform normalization

Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables, and put $\pi_j = \binom{n}{j} p^j q^{n-j}$, zero outside the legal counts. Define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}, \quad O = \mathbb{P}(W_k > u) + f(u), \quad T = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i).$$

Conditioning on the sign count, including its atoms, gives

$$Q := \mathbb{P}(S_n \le H) - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i} \mathbb{P}(W_{k-i} > u + i) + \sum_{i=1}^{n-k} \pi_{k+i} \mathbb{P}(W_{k+i} \le u - i).$$

Consequently, if $\pi_{k+i}/\pi_k \le R_0$,

$$a + \rho_B(C_{\mathrm{E}} - C_B) \le -\sqrt{n}\pi_k(O - R_0 T) + \mathcal{R}, \quad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C_{\mathrm{E}}(\rho_B - \rho). \tag{1}$$

The discarded terms are nonpositive. The outgoing event is strict and the incoming events closed, also at a collision.

Here $v \ge 6/25$ and $\sqrt{v+s} < 501/1000$. Thus $|\delta| < 33/20$ throughout; on $|u| \le 1/20$ it is less than one. These inequalities imply the stated interior-count range and $9/25 < k/n < 23/40$. On this last interval the absolute third derivative of binary relative entropy is at most 6, the absolute logarithmic derivative of $r(1-r)$ at most $5/4$, and $1/(12r(1-r)) < 3/8$. The finite Stirling estimate therefore gives

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| < Z(D) := 5D/8 + 3/8 + D^3/42$$

whenever $|\delta| \le D \le 33/20$. This follows by expanding relative entropy to second order and the logarithmic prefactor to first order; the three remainders are respectively D^3/n^2, $5D/(8n)$ and $3/(8n)$. It uses the same finite Stirling formula proved in the manuscript's count appendix, not an asymptotic law. Using $0.39886 < \phi(0) < 0.4$, $e^{-x} \ge 1 - x$, and $e^x \le (1-x)^{-1}$ for $x < 1$ yields the following bounds:

	$\lvert u\rvert \le 1/20$	$-34/125 \le u \le 91/125$
$\sqrt{n}\pi_k$ lower bound	$73/100$	$13/20$
f' lower bound	$19/20$	$4/5$
f' upper bound	$21/20$	$5/4$
π_{k+i}/π_k upper bound	$53/50$	$3/2$
$\mathcal{R}$ upper bound	$2\lambda/n$	$3\lambda/n$

In the derivative bounds the actual δ is held fixed and the derivative variable runs between zero and the actual phase. Indeed

$$\log f'(t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+t)^2}{2n(v+s)} - e,$$

and, for $|t| \le b$,

$$n|\log f'(t)| \le (8/3)\lambda + \frac{2Db + b^2}{2(6/25)} + Z(D).$$

Take $(D, b) = (1, 1/20)$ or $(33/20, 91/125)$. At a binomial mode, $|\delta| \le 1$; the same Stirling estimate shows $\sqrt{n}\max_j \pi_j < 17/20$. Dividing this by $13/20$ proves the global ratio bound.

In the inner phase range, the exact binomial recursion gives $\pi_{k+1}/\pi_k \le 1 + (53/100)/(42 \cdot 6/25) < 53/50$; all subsequent factors are at most one since $(n+1)p - k < 153/100 < 2$.

For the repair bound, conditional signs are fixed because $a_0 < p$ and $a_1 < q$. Expansion of the absolute cubic gives, with $r = \min(p, q)$,

$$\mathbb{E}|Y - p|^3 \ge v(1-2v) + (3r - d)s, \qquad d = a_0 + a_1.$$

This follows term by term from $U_1^3 \ge -a_1 U_1^2$ and $U_0^3 \le a_0 U_0^2$. Since $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$, it follows that

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\{3r - \tfrac{3}{2}(1-2v) - d - 3(1-2v)s/(8v)\} > 0.$$

The bracket is positive using $r \ge 2/5$, $v \ge 6/25$ and $s \le 1/21000$. We may therefore discard the favorable moment term. Differentiating the normal term in s bounds the remaining repair by $(7/4)\delta_+ s$, because $0.2/(6/25)^{3/2} < 7/4$. This proves the two repair bounds in the table.

1.2 Incoming counts need only a fourth moment

For a centered independent sum W whose summands have absolute bound $A = 3/5$ and total variance t,

$$\mathbb{E}W^4 \le A^2 t + 3t^2.$$

Also $|s_1 - s_0| \le (5/2)s$, so that $t_{k+i} \le iV\lambda$, with $V = 9/8$ in the inner phase range and $V = 7/6$ globally. Markov's inequality, applied to a closed incoming event, therefore gives, for any phase upper bound $h \in [0, 1)$,

$$\frac{T}{\lambda} \le \frac{K(V)}{(1-h)^4}, \qquad K(V) = \frac{9}{25} V \frac{49}{40} + 3V^2 \frac{1}{500}\frac{17}{10}. \tag{2}$$

Indeed $i - u \ge i(1-h)$; sum the resulting i^{-3} and i^{-2} terms. The elementary integral bounds

$$\sum_{i \ge 1} i^{-3} \le 1 + 1/8 + 1/27 + 1/18 < 49/40, \quad \sum_{i \ge 1} i^{-2} \le 1 + 1/4 + 1/9 + 1/3 < 17/10$$

justify these constants. For $u \le 0$ use $h = 0$.

1.3 The own count, uniformly down to zero noise

Set $\alpha = 9/20$ and

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x}{5\alpha^4}.$$

Then $h_\alpha \le \mathbf{1}_{(0,\infty)}$: the polynomial is nonpositive on the negative half-line and its positive maximum is $(9 + 6\sqrt{3})/20 < 1$. Its value at zero is zero. For a centered independent sum W as above, Cauchy and the moment expansion give $|\mathbb{E}W^3| \le At$ and $\mathbb{E}W^4 \le A^2 t + 3t^2$. Thus

$$\mathbb{P}(W + \sigma r > 0) \ge \frac{(3\alpha^2 - A^2)t - 3t^2 - 4Art - 6r^2 t - r^4 + 3\alpha^2 r^2 + 2\sigma\alpha^3 r}{5\alpha^4}.$$

In the inner phase range, $79\lambda/84 \le t = \mathbb{E}W_k^2 \le 89\lambda/84$. Put $T_* = 89/42000$. To bound the preceding expression plus br, divide by $t > 0$ and write $r = zt$. The result is

$$F_{\sigma,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{5\alpha^4} + \left(\frac{2\sigma}{5\alpha} + b - \frac{4At}{5\alpha^4}\right) z + \frac{3\alpha^2 t - 6t^2}{5\alpha^4} z^2 - \frac{t^3 z^4}{5\alpha^4}.$$

It is concave in $t \geq 0$. On $0 \leq z \leq z_0$, replace z^4 by $z_0^2 z^2$ to obtain a lower quadratic. The minimum of this quadratic on the interval occurs at an endpoint or its eligible vertex. Checking it at $t = 0, T_*$ proves

$$F_{1,-21/20}(z,t) > 4/5 \quad (0 \leq z \leq 2), \qquad F_{-1,19/20}(z,t) > 1 \quad (0 \leq z \leq 3/2).$$

The respective rational margins are greater than 0.0288 and 0.176. These are interval inequalities, not sampled noise checks.

For negative phase $u = -r$ and $r \geq 2t$, Cantelli instead gives $O \geq r^2/(t + r^2) - (21/20)r$. At $r = 2t$ its value divided by t exceeds $4/5$, and its derivative in r is positive. That derivative first increases and then decreases, so any subsequent stationary point is a maximum. The value at the other endpoint $r = 1/20$ also exceeds $(4/5)t$, checked at $t = T_*$. Hence $O > (4/5)t$ throughout the negative inner phase range. For positive phase, the quartic handles $r \leq 3t/2$; for larger r, simply use $O \geq f(r) \geq (19/20)r > t$. Both switching points are below $1/20$. We have proved

$$O > \frac{4}{5}\frac{79}{84}\lambda \qquad (|u| \leq 1/20).$$

Combining this with (2) and the table yields

$$\frac{73}{100}\left\{\frac{4}{5}\frac{79}{84} - \frac{53}{50}\frac{K(9/8)}{(19/20)^4}\right\} - \frac{2}{42} - \frac{1}{100} = \frac{219609181}{27367410000} > 0.$$

The bracket is positive before the lower mass bound is used in (1).

1.4 The two outer phase intervals

For $u = -r \in [-34/125, -1/20]$, Cantelli and the global derivative bound give

$$O \geq \frac{r^2}{11/5000 + r^2} - \frac{5}{4}r.$$

This expression is concave on the stated interval because $3r^2 > 11/5000$. Its smaller endpoint value exceeds 0.469. Subtract $(3/2)K(7/6)/500$ for incoming counts. Multiplication by the mass lower bound $13/20$ still exceeds the repair $3/(42 \cdot 500)$ and the desired $1/50000$ margin.

For $u \in [1/20, 91/125]$ use only $O \geq (4/5)u$. The function $u(1-u)^4$ increases and then decreases on $[0,1]$; its minimum on this interval is $G = (91/125)(34/125)^4$. Consequently

$$O - R_0 T \geq (4/5)G - (3/2)K(7/6)/500 > 0.$$

Here the positive numerator was divided by $(1-u)^4 \leq 1$ before dropping that denominator. Finally

$$\frac{13}{20}\{(4/5)G - (3/2)K(7/6)/500\} - \frac{3}{42 \cdot 500} - \frac{1}{50000} = \frac{4806449287359}{5468750000000000} > 0.$$

Using the upper noise endpoint is valid in both outer intervals since every discarded error increases with λ. This completes the proof of Theorem 1.1.

The accompanying exact rational checker reconstructs all 46 scalar gates. Its rejection of the same constants at $\lambda = 1/20$ is a negative control for the sufficient argument, not a counterexample to the theorem or to the Berry–Esseen conjecture. The range $1/500 < \lambda$ still needs a different estimate or sharper constants.